

MATH 5440

Mathematics of Coding and Cryptography

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	Machine Learning Engineer (23.5 / 100)	15.1 / 100

15

CURRENT AVG SCORE
across 6 job roles

62

POTENTIAL SCORE
with recommended additions

+62

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Gives an introduction, with proofs, to the algebra and number theory used in coding and cryptography. Basic problems of coding and cryptography are discussed; prepares students for the more advanced ECEN 5682 .

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	23.5	66.2	+42.7	HIGH
AI Engineer	22.6	62.0	+39.4	HIGH
Data Scientist	16.5	66.8	+50.3	HIGH
ML Engineer	11.6	63.1	+51.5	HIGH
Applied Scientist	9.0	42.0	+33.0	HIGH
MLOps Engineer	7.3	69.7	+62.4	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (example) appear as gaps across 3 or more job roles.

System Design

4 of 6 roles

systems

Business Communication

1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **HIGH** **Add System Design coverage** — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 **LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

Machine Learning Engineer

23.5 → 66.2 (+42.7 pts)

data learning tools protection business design science models

AI Engineer

22.6 → 62.0 (+39.4 pts)

data models learning build software code systems based

Data Scientist

16.5 → 66.8 (+50.3 pts)

data product presentation different build approaches

ML Engineer

11.6 → 63.1 (+51.5 pts)

data learning systems models build impact product digital

Applied Scientist

9.0 → 42.0 (+33.0 pts)

computational models data learning systems design studies fundamental

MLOps Engineer

7.3 → 69.7 (+62.4 pts)

detection data scientists text systems learning image tools

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.